

Online Fake News Detection using Convolutional Neural Network and Transformers

Team Name: News Detectives

Project Proposal

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1 INTRODUCTION

False information and fake news fundamentally impact our daily lives on economic, political, and social scales. According to [2], news delivered through social media has been a focus of concern. One study suggests that 62 percent of US adults get their news from social media, and more than 40 percent of them do so on Facebook. Estimates suggest that in the final three months of the 2016 US presidential campaign, the top-performing fake election news stories on Facebook attracted more views than top stories from major news outlets such as the New York Times, Washington Post, Huffington Post, or NBC News. And with the rise of such events, the field of fake news detection also made significant progress, especially in utilizing several deep neural network techniques to detect these fake news successfully.

Our research will leverage the comprehensive "Fake News" dataset available on Kaggle, as it comprises a collection of news stories with their titles, authors, countries, images, and labels such as "BS", "bias", "fake", and "conspiracy". This dataset will allow us to train and test various deep-learning models to identify the underlying patterns and features that differentiate fake news from legitimate reports.

The potential impact of this study is substantial, as the study could lead to practical solutions to critical issues of fake news, which may affect the worldwide information policy generation and interventions and reduce the destructive effects of false information on trusted media. We kindly request your consideration of our proposal and look forward to the opportunity to contribute to the scientific community's understanding of this urgent issue.

2 APPROACH

We intend to explore various deep learning algorithms, including but not limited to sequence-to-sequence transformers and convolutional neural networks, to enhance our fake news detection performance. We will consider utilizing sentiment analysis techniques with transformers like Bidirectional Encoder Representations from Transformers (BERT) and Robustly Optimized BERT-Pretraining Approach (RoBERTa). To improve the models' performance, we will fine-tune these models using fully connected layers and max pooling. For optimization, we will employ the "Adam" optimizer and utilize cross-entropy loss as the loss function. The evaluation of our models will involve measuring metrics such as accuracy, precision, recall, and F1 score.

Furthermore, we will investigate different feature extraction techniques, such as TF-IDF and word embeddings, to capture the semantic and syntactic structure of the news title content. By analyzing miss-classified instances, we aim to understand the limitations and complexities of fake news generation. This analysis will provide valuable insights for further system enhancement and a deeper understanding of fake news characteristics. Finally, we might be able to validate the final model across dataset [3] to test the performance of the model.

3 RESOURCES, RELATED WORK & PAPERS

The field of fake news detection has made significant progress, and the most effective methods now use a

combination of techniques. These techniques are discussed in the following papers:

1. Detection of Online Fake News Using N-Gram Analysis and Machine Learning Techniques [1]. A model that uses n-gram analysis and machine learning techniques to detect fake news was proposed in this paper. TF-IDF and Linear Support Vector Machine were used for feature extraction and classification, respectively, with an accuracy of 92%.

2. Detection of fake news using deep learning CNN-RNN based methods [4]. This study focuses on detecting fake news using deep learning techniques like Bidirectional LSTM, CNN, and ResNet, combined with pre-trained word embeddings. Data augmentation through back-translation is employed to address class imbalances. Bidirectional LSTM demonstrated superior performance compared to CNN and ResNet across all datasets.

3. Fake News Detection on Social Media: A Data Mining Perspective [5]. Fake news on social media can have a big negative impact, and it can be difficult to detect. This survey looks at different ways to detect fake news on social media, including using different algorithms, evaluation metrics, and datasets. The survey also discusses future research directions, emphasizing the need to use auxiliary information and address the complexities of unstructured and noisy user engagement data.

4. Detecting fake news by exploring the consistency of multimodal data [6]. A new method for detecting fake news during the (2019-nCoV) outbreak uses a multimodal neural network to extract text and visual features, measure similarity, and fuse the results. The method outperforms existing approaches on four datasets.

5. Hong Kong Protests: Using Natural Language Processing for Fake News Detection on Twitter [7]. This paper automates fake news detection in politically-oriented news related to the Hong Kong protests using Twitter data. By employing language-independent filtering and machine learning algorithms, it identifies linguistic differences between fake and real news tweets, which aligns with previous studies conducted in this area.

6. Fake news detection: A hybrid CNN-RNN based deep learning approach [3]. This paper proposes a novel hybrid deep learning model that combines convolutional and recurrent neural networks for fake news classification. The model was successfully validated on two fake news datasets (ISO and FA-KES), achieving detection results that are significantly better than other non-hybrid baseline methods. Further experiments on the generalization of the proposed model across different datasets, had promising results.

4 DATASETS

1. Original dataset from Kaggle: [Fake and real news dataset](#).

2. Fake news dataset organized: [News Articles dataset](#).

3. Fake news dataset from Kaggle for performance validation: [Fake News Dataset around the Syrian War](#).

5 TEAM MEMBERS

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